

Using rates 1A

1

James is paid \$13.70 per hour.
How much is he paid for 7 hours' work?

Multiply the rate per hour by the number of hours:

$$13.70 \times 7$$

James is paid \$95.90

2

Tomatoes were priced at \$5.90 per kilogram.
How much would 2.3 kg of tomatoes cost?

Multiply the rate per kilogram by
the number of kilograms:

$$\times 2.3$$

2.3 kg of tomatoes cost \$13.57

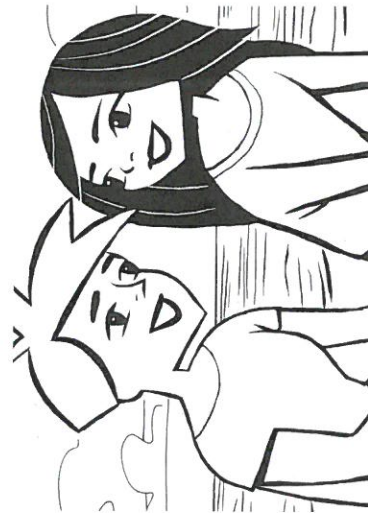
3

The instructions for a brand of fertiliser state that
1 kg should be applied to an area of 25 m².
(The rate of cover is 25 m²/kg.)
What area can be covered using 15 kg of this fertiliser?

Multiply the rate by the number of kilograms:

$$\times 15$$

15 kg of fertiliser will
cover 375 m²



4

Rhani is paid \$14.50 per hour.
Calculate how much she earns when she works:

A 4 hours. \$58

B 5 hours. \$72.50

C 12 hours. \$174

D 20 hours. \$290

5

A tiler lays 2 m² of tiles per hour.
Find the number of square metres she can lay in:

A 3 hours. 6 m²

B 5 hours. 10 m²

C 1 $\frac{1}{2}$ hours. 3 m²

D 3 $\frac{1}{2}$ hours. 7 m²

Using rates 1B

RATES

UNIT 1
TOPIC 2

- 6** A cricket team scored an average of 5 runs per over.
How many runs did they score off 19 overs?

95 runs

- 7** Rochelle can shear 14 sheep per hour.
How many sheep could she shear in 8 hours?

112

- 8** A farmer can have 10 sheep per hectare on his property. He has 75 hectares of land for sheep.
How many sheep can he have?

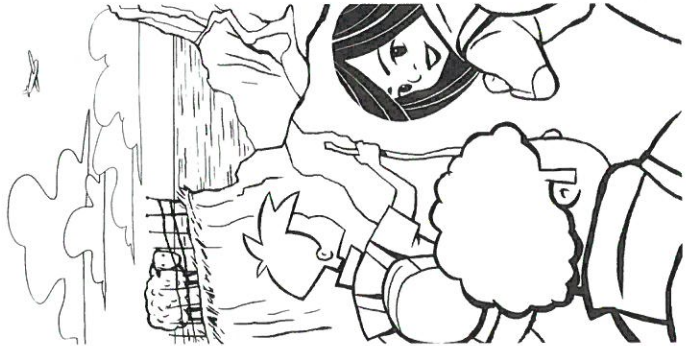
750

- 9** An aeroplane can carry 135 passengers per flight. Yesterday, it flew 8 full trips. How many passengers did it carry?

1080

- 10** A heart is beating at 70 beats per minute.
How many beats will it make in 1 hour? (1 hour = 60 minutes)

4200



- 11** Water is dripping from a tap at a rate of 4 litres per hour.
How much water will drip from the tap in 1 day?
(1 day = 24 hours)

96L

- 12** Marko estimates that his feed supply will sustain 5 head of cattle per hectare.
How many head of cattle could he put on an area of 250 hectares?

1250

- 13** A mine supplies ore at a rate of 1 600 tonnes per day. The mine operates 7 days each week.
How many tonnes of ore can it supply in 3 weeks?

33600

- 14** The temperature at 8:00 am was 15°C .
If it increased at a steady rate of 2°C per hour, what was the temperature at:

- | | | |
|----------|-----------|------|
| A | 9:00 am? | 17°C |
| B | 11:00 am? | 21°C |
| C | 12 noon? | 23°C |
| D | 10:30 am? | 20°C |
| E | 12:30 pm? | 24°C |



Using rates 2B

- 9** A piece of fabric is 3 metres long. It costs \$48.

A What is the cost per metre? \$16.

B What would be the cost of a piece measuring 5 metres long?

\$80

C What length of fabric can be purchased for \$56?

3.5m

- 10** Tyrone sheared 34 sheep in 2 hours. If he continues working at the same rate:

A what is Tyrone's shearing rate in sheep per hour?

17 sheep/h

B how long will it take him to shear 85 sheep?

5

- 11** Phuong can type at a rate of 75 words per minute.

How long will it take him to type a 1 500 word document at this rate?

20mins.



- 12** A factory produces drink cans at a rate of 200 cans per minute.
How long would it take to produce 1 500 cans?

7.5 mins

- 13** Martin serves an average of 24 customers every 2 hours.
At this rate, how long would it take him to serve 54 customers?

4.5 hours

- 14** A team of bricklayers takes 6 hours to lay 1 500 bricks.
Working at the same rate, how long will it take them to lay 2 500 bricks?

10 hours.

- 15** The annual birth rate of a population is usually expressed as the number of births each year per 1 000 people.

A country has an annual birth rate of 24 per 1 000 people.
Calculate the number of births expected next year in a town of:

A 5 000 people. 120

B 30 000 people. 720

C 85 000 people. 2040

D 165 000 people. 3960

E 128 500 people. 3084.



Using rates 2A

Many problems using rates can be solved using these steps:

- Step 1: Write down the rate.

Step 2: Reduce the rate to 1 unit (by division).

Step 3: Multiply or divide to get the answer.

To decide whether to multiply or divide, check the position of the required quantity in the rate.

- To find the first quantity you multiply.
- To find the second quantity, you divide.

1 Eleni delivers 320 letters in 4 hours.

A What is her delivery rate per hour? 80 letter/h.

B How many letters can she deliver in 5 hours?

400

2 David earns \$170.25 for 15 hours' work.

A How much is he paid per hour?

\$11.35

B How much would he earn for 7 hours' work?

\$79.45.



RATES

UNIT 1
TOPIC 2

You may need to use working paper to answer these questions.

3 Sarah delivered 28 parcels in 4 hours.

At this rate, how many parcels would she deliver in 5 hours?

35

4 A truck can carry 182 tonnes in 7 trips.

How many tonnes can it carry in 5 trips?

130

5 On a sheep farm, 350 sheep can graze on 5 hectares of pasture.

How many sheep can graze on 16 hectares?

1120

6 A wheel spins at 45 revolutions per minute.

How many revolutions will it make in 1 hour?

2700

7 A factory fills soft drink bottles at a rate of 80 bottles per minute.

How many bottles can be filled in 2 hours?

9600

8 A pump can deliver 55 litres of water in 5 minutes.

How many litres can it deliver in 3 hours?

1980

